

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element H Initial Template

How to use this template

Guidance before filling out the table on the next page:

- Enter *all* of the high-priority design requirements you listed in Element C into the table below.
- Determine an appropriate testing method for each requirement. For example, the test may involve measurement, inspection, physical actions, or mathematical modeling.
- As you plan, use your scientific method best practices, e.g. creating experimental controls, defining independent and dependent variables, only changing one variable at a time and changing it in increments to find a trend, state your prediction, etc.
- Your testing plan for a given requirement must include enough detail so that a person who is not on the design team could fully replicate the test. Include photos and/or diagrams to add clarity. Each test should include the following details:
 - how the test will be set up
 - what needs to happen during the test
 - test cases (if applicable)
 - number of trials of the test expected (aim for 7 or more trials)
 - the expected range of results that could be recorded, including units
 - the criteria for passing or failing the associated design requirement, including any permissible range for error and/or minimal thresholds
 - any goals your team has set beyond the passing result

Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1 ex	The prototype must remain intact when held upside down.	1) Hold the prototype in a hand using the grip in this photo. <insert grip photo> 2) Take 3 seconds to slowly rotate the prototype around the indicated axis into an upside down position with a smooth motion. <insert photo with annotation of axes of rotation> 3) Note any observations about prototype integrity. 4) Take 3 seconds to rotate the prototype in reverse, back to its original position, using a smooth motion	a. Rotate around x-axis (10 trials) b. Rotate around y-axis (10 trials) c. Rotate around z-axis (10 trials)	Parts of the prototype may: – remain unchanged – deform – shift position – tear – detach	Fail: one or more parts of the prototype tears or detaches Pass: zero parts of the prototype tear or detach Goal: all parts of the prototype remain unchanged when the prototype is returned to its original position
2 ex	mass of prototype < 3kg	1) Tare a digital scale. 2) Place the prototype securely on the scale. 3) Wait for the reading on the scale to stabilize before recording the value.	test cases: N/A # trials: 10	0kg<reading<10kg	Fail: mean of the trials $\geq 3\text{kg}$ Pass: mean of the trials < 3kg Goal: mean of the trials < 2kg
3	ROV should be capable of movement in	1) Place the ROV into the hot tub 2) Let ROV fully submerge with the assistance of a downwards-moving	#2 trials	ROVs movement - Is controlled - Speed is	Fail: water was inside the ROV with no way for it to

	directions of a 360-degree radius. As well as move in a straight line just below the surface.	motor. 3) Drive the ROV in a straight line, then rotate the ROV 180° and move in a straight line from that position. 4) move in a squiggly line to ensure that the movement from both sides is possible and controlled.		maintained - Able to rotate - And move in a straight line	return to the surface, and it needed to drill holes for water to escape. Pass: was able to move in a 360° and pass the straight line tests, after the editing of the ROV itself. Goal: all movement is possible
4	ROV should be capable of buoyancy and resurfacing after emerging underneath the site of interest	1) Place the ROV into the water 2) Let the ROV fully submerge with assistance from the the motors 3) Go down as far as possible, then go up to return to the surface.	5 trials	ROV can go down and submerge with no issues, then resurface with no issues.	Fail: the ROV was able to submerge itself, but when water go into the tubing there was no way for the water to escape, so it became very heavy and found it difficult to resurface. Pass: after editing the ROV by drilling holes so that water can escape and not get trapped, the ROV successfully

					resurfaced.
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Feedback From Field Experts (H5)

Guidance to students about this section of Element H.

- a) "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.
- b) You must provide evidence in this section that your testing plan has been reviewed by 2-3 field experts.
- c) Do not incorporate their feedback in the table above. Rather, make a copy of the table in the section above, paste it into the section below, and make all revisions only in the table below. This provides evidence that you incorporated the feedback and shows your testing plan's progression.

Revised Testing Details That Incorporate Field Expert Feedback (H5)